Generalized Stokes' Theorem

An End-to-End Meta-Math Walkthrough

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Abstract

A complete authored walkthrough of generalized stokes' theorem, with explicit mathematical contracts, local verification gates, and reusable exports. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

AC-05 / Advanced Calculus / Report v1

I. TARGET AND IMPORT CONTRACT

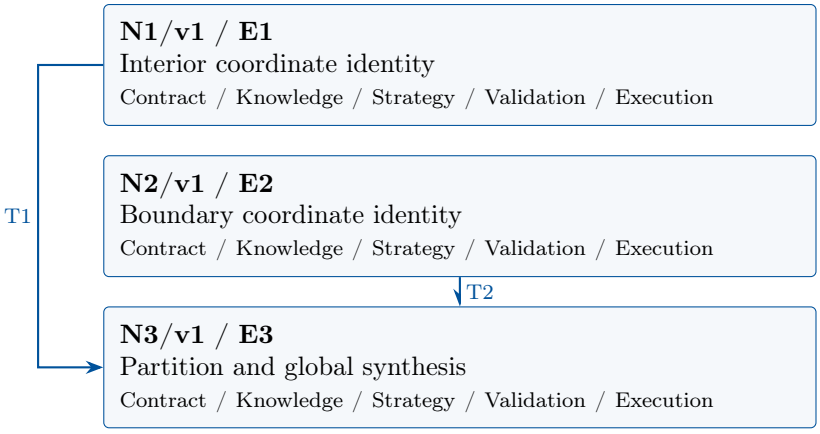
Theorem I.1. *For a compact oriented smooth n-manifold M with boundary and a smooth $(n-1)$-form ω, prove $\int_M d\omega = \int_{\partial M} \omega$, with the outward-normal-first boundary orientation.*

A. Foundations and permitted imports

Import oriented smooth charts (including half-space charts), smooth partitions of unity subordinate to a finite chart refinement of compact M, coordinate invariance of integration of forms, and AC-02. A partition can be chosen with each localized support compactly contained in its chart relative to the half-space. For $n = 1$ the boundary integral of a0-form is the signed endpoint sum; $n \geq 1$ is intended. Boundary orientation is outward-normal-first.

Fubini for continuous compactly supported functions is also used in each coordinate chart. The dimension satisfies $n \geq 1$.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1: N1/E1/v1 to N3. Stokes' identity for compactly supported interior-chart forms.

Receiving activity. N3: Partition and global synthesis

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Finite partition ρ_j with $\sum \rho_j = 1$; both coordinate identities; prior exports E1/v1, E2/v1

T2: N2/E2/v1 to N3. Stokes' identity in an oriented boundary chart with the required sign.

Receiving activity. N3: Partition and global synthesis

Compatibility. Use the same mathematical objects and hypotheses as the source export. Receiving contract: Finite partition ρ_j with $\sum \rho_j = 1$; both coordinate identities; prior exports E1/v1, E2/v1

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Interior coordinate identity

Object and role. Integrate each partial derivative over its coordinate.

Inputs. Compactly *supported*$(n-1)$-form in oriented R^n; foundational imports stated in §1

C — Contract

Own E1/v1: Stokes' identity for compactly supported interior-chart forms.

K — Knowledge

Retain Compactly *supported*$(n-1)$-form in oriented R^n; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Integrate each partial derivative over its coordinate.

V — Validation

Compact support removes artificial chart-edge boundary terms.

E — Execution

Carry out the numbered deductions and publish E1/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N1.1 Write $\eta = \sum_{i=1}^{n} (-1)^{i-1} a_i \; dx_1 \wedge \cdots \wedge omit \; (dx_i) \wedge \cdots \wedge dx_n$.

N1.2 Then $d\eta = (\sum_i \partial_i \; a_i) dx_1 \wedge \cdots \wedge dx_n$.

N1.3 Fubini and FTC imply each full-space integral of $\partial_i \; a_i$ is 0, since a_i has compact support.

N1.4 Thus $\int d\eta = 0$, matching the empty boundary contribution of an interior chart.

Export E1/v1. Stokes' identity for compactly supported interior-chart forms.

Acceptance gate. Compact support removes artificial chart-edge boundary terms. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N1: Compact support removes artificial chart-edge boundary terms. Recheck dependent nodes N3.

B. N2/v1: Boundary coordinate identity

Object and role. Track the outward-first orientation explicitly.

Inputs. The same form on half-space $x_n \geq 0$; foundational imports stated in §1

C — Contract

Own E2/v1: Stokes' identity in an oriented boundary chart with the required sign.

K — Knowledge

Retain The same form on half-space $x_n \geq 0$; foundational imports stated in §1; keep their assumptions and versions attached.

S — Strategy

Track the outward-first orientation explicitly.

V — Validation

A chart boundary is not automatically positively oriented in tangent coordinates.

E — Execution

Carry out the numbered deductions and publish E2/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N2.1 Tangential derivative integrals vanish as before.

N2.2 The normal term gives $\int_0^\infty \partial_n\ a_n\ dx_n = -a_n(x', 0)$, so $\int d\eta = -\int a_n(x', 0)dx'$.

N2.3 The outward normal is $-\partial_n$.

N2.4 Relative to the standard tangent frame $(\partial_1, \ldots, \partial_{n-1})$, the outward-first frame has orientation $\mathrm{sign}(-1)^n$.

N2.5 The restriction of η is $(-1)^{(n-1)} a_n\ dx_1 \wedge \cdots \wedge dx_{n-1}$.

N2.6 Multiplying these signs gives $-a_n$, exactly the computed boundary integral.

N2.7 For $n = 1$ this says the boundary at 0 contributes $-a_1(0)$.

Export E2/v1. Stokes' identity in an oriented boundary chart with the required sign.

Acceptance gate. A chart boundary is not automatically positively oriented in tangent coordinates. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N2: A chart boundary is not automatically positively oriented in tangent coordinates. Recheck dependent nodes N3.

C. N3/v1: Partition and global synthesis

Object and role. Localize the form, then sum without adding spurious derivatives.

Inputs. Finite partition ρ_j with $\sum \rho_j = 1$; both coordinate identities; prior exports E1/v1, E2/v1

C — Contract

Own E3/v1: $\int_M d\omega = \int_\partial M\omega$ for the stated compact oriented manifold.

K — Knowledge

Retain Finite partition ρ_j with $\sum \rho_j = 1$; both coordinate identities; prior exports E1/v1, E2/v1; keep their assumptions and versions attached.

S — Strategy

Localize the form, then sum without adding spurious derivatives.

V — Validation

The localized derivative is $d(\rho_j\omega)$, not merely $\rho_j d\omega$.

E — Execution

Carry out the numbered deductions and publish E3/v1 together with its hypotheses and proof location.

Numbered mathematical activities

N3.1 Write $\omega = \sum_j \rho_j\omega$.

N3.2 Each localized form has support within its chart, so the appropriate identity applies after pullback and extension by zero away from the chart edges.

N3.3 Linearity gives $\int_M d\omega = \sum_j \int_M d(\rho_j\omega) = \sum_j \int_\partial M\ \rho_j\omega = \int_\partial M\omega$.

N3.4 Equivalently the extra $d\rho_j \wedge \omega$ terms sum to $d(\sum \rho_j) \wedge \omega = 0$.

N3.5 Compactness makes the sum finite.

N3.6 This proves the global formula with no unexamined chart-interface terms.

Export E3/v1. $\int_M d\omega = \int_\partial M\omega$ for the stated compact oriented manifold.

Acceptance gate. The localized derivative is $d(\rho_j\omega)$, not merely $\rho_j d\omega$. Publish only after checking every displayed deduction.

Targeted repair. If the check fails, revise N3: The localized derivative is $d(\rho_j\omega)$, not merely $\rho_j d\omega$. Withhold this terminal result until its gate passes.

IV. CONTINUOUS MATHEMATICAL PROOF

Read these steps in order as one proof; the diagram is optional. Numbered activities identify the specific constructions and deductions owned by each node.

A. §4.N1 — Interior coordinate identity

1. Write $\eta = \sum_{i=1}^{n} (-1)^{i-1} a_i \; dx_1 \wedge \cdots \wedge omit \; (dx_i) \wedge \cdots \wedge dx_n$.
2. Then $d\eta = (\sum_i \partial_i \; a_i) dx_1 \wedge \cdots \wedge dx_n$.
3. Fubini and FTC imply each full-space integral of $\partial_i \; a_i$ is 0, since a_i has compact support.
4. Thus $\int d\eta = 0$, matching the empty boundary contribution of an interior chart.

B. §4.N2 — Boundary coordinate identity

1. Tangential derivative integrals vanish as before.
2. The normal term gives $\int_0^\infty \partial_n \; a_n \; dx_n = -a_n(x', 0)$, so $\int d\eta = -\int a_n(x', 0) dx'$.
3. The outward normal is $-\partial_n$.
4. Relative to the standard tangent frame $(\partial_1, \ldots, \partial_{n-1})$, the outward-first frame has orientation $\text{sign}(-1)^n$.
5. The restriction of η is $(-1)^{(n-1)} a_n \; dx_1 \wedge \cdots \wedge dx_{n-1}$.
6. Multiplying these signs gives $-a_n$, exactly the computed boundary integral.
7. For $n = 1$ this says the boundary at 0 contributes $-a_1(0)$.

C. §4.N3 — Partition and global synthesis

1. Write $\omega = \sum_j \rho_j \omega$.
2. Each localized form has support within its chart, so the appropriate identity applies after pullback and extension by zero away from the chart edges.
3. Linearity gives $\int_M \; d\omega = \sum_j \int_M \; d(\rho_j \omega) = \sum_j \int_\partial M \; \rho_j \omega = \int_\partial M \omega$.
4. Equivalently the extra $d\rho_j \wedge \omega$ terms sum to $d(\sum \rho_j) \wedge \omega = 0$.
5. Compactness makes the sum finite.
6. This proves the global formula with no unexamined chart-interface terms.

□

V. WORKED ORCHESTRATION

On the unit disk in R^2 with orientation $dx \wedge dy$, take $\omega = (-y\ \mathrm{dx}{+}\mathrm{x}\ \mathrm{dy})/2$. Then $d\omega = dx \wedge dy$, whose integral is π. On the counterclockwise circle $x = cost, y = sint$, $\omega = dt/2$ and the boundary integral is π, exercising the orientation convention.

A. Boundary and hypothesis audit

For the interval[0,1], Stokes gives $\int f'dx = f(1) - f(0)$; treating both endpoint orientations as positive would give the wrong result. Validation returns this one-dimensional sign test before accepting the boundary-chart export.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Failure and correction

This is an authored teaching trace, not a recorded agent run.

1. **Contract → Knowledge:** register the §1 theorem, imports, and requested exports before beginning the proof.
2. **Strategy → Execution:** choose the bridges in §3 and carry out the numbered activities in §4; transmit only established E-exports along the §2 DAG.
3. **Validation → responsible node:** test the specific boundary/failure in §5 and the node's individual audit. A failing candidate is withheld; downstream nodes may not use it.
4. **Knowledge + Strategy:** identify whether the error is an omitted condition, an invalid inference, or an attempted extension beyond the theorem. Restore the exact contract or explicitly state a different one.
5. **Execution:** redo the affected mathematical activity using the repaired condition or inference. Recheck consuming nodes, rather than feeding their conclusions back as premises.
6. **Contract:** publish only the supported export with its assumptions and proof location. A hypothetical revised export is v2; the written proof here is the v1 exposition, not a claim that this repair was executed.

B. Hypothesis boundary

For the interval[0,1], Stokes gives $\int f'dx = f(1) - f(0)$; treating both endpoint orientations as positive would give the wrong result. Validation returns this one-dimensional sign test before accepting the boundary-chart export.

C. Version and recheck

Treat any changed assumption as a new contract. Recheck the revised export and all of its descendants. A backward diagnostic never establishes an earlier theorem. This is an authored teaching trace, not an execution log.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading.

- Harvard Math 230a: Riemannian geometry, curvature, and Hopf–Rinow.